



Edward Tatum. He and his colleague George W. Beadle developed the "one gene-one enzyme" model of gene function.

**phenotype** observable characteristics of an organism

By the time of Mendel's rediscovery in 1900 by Hugo de Vries, Carl Correns, and Erich Tschermak, however, visible characters were understood to be expressions of hereditary particles within each cell. Just as characters occurred in pairs, Mendelians interpreted hereditary particles as occurring in pairs. William Bateson called these pairs of hereditary particles "allelomorphs," a term that would eventually be shortened to "alleles." The idea that something within the gametes (sperm and egg cells) specified the characteristics of the organism was captured by Johannsen's term "gene."

It was not at all clear in the first decade of the twentieth century that the segregation and assortment of alleles could explain patterns of inheritance. While Bateson forcefully advocated Mendelian principles, many of his contemporaries, such as Karl Pearson and Walter F. R. Weldon, explained patterns of heredity in terms of continuous characters, instead of discrete Mendelian character pairs. For Pearson, Weldon, and other biometricians, as they called themselves, traits were expressed as continuous distributions from tall to short, for instance. The mating of a tall parent with a short parent would yield offspring with a range of heights, as the parental traits were blended together.

Because Mendelians, like Bateson, strongly identified characters with alleles, they insisted that both were discrete. The apparent blending of parental traits was a stumbling block until the distinction between genotype and **phenotype** was consistently applied and single continuous traits such as height were seen to be the expression of many individual Mendelian factors. Thus, multifactor inheritance allowed geneticists to explain a continuously distributed phenotype as an expression of many discrete Mendelian factors or genes.

## The Chromosome Theory

Of the many difficulties facing early genetics, one of the most important was figuring out what a gene was actually made of. As early as 1902, Walter Sutton and Theodore Boveri had observed that, during meiosis, chromosomes separated just as Mendelian particles were proposed to separate. The discovery of sex chromosomes a few years later suggested that chromosomes might play a genetic role, but the association of a specific gene with a chromosome would not occur until 1910, when Thomas Hunt Morgan demonstrated the sex-limited inheritance of the white-eye mutation in *Drosophila*.

Using the wealth of new information provided by their experiments with *Drosophila*, Morgan and his colleagues articulated the chromosome theory of inheritance, which treated genes as indivisible particles arranged like beads on a string to form a chromosome. Their patterns of association provided the clues to map their linear order on chromosomes and to understand processes of chromosomal recombination and rearrangement.

## Gene Action and Mutation

The tremendous successes of the Morgan group often overshadow a simultaneous tradition of exploring the nature of gene action. The problem of how genes produce their effects was the domain of physiological genetics. As early as 1911, A. L. and A. C. Hagedoorn had proposed that genes acted as chemical catalysts. In 1916 Richard Goldschmidt interpreted genes as

**enzymes**, while Sewall Wright explained coat-color patterns in terms of genetic regulation of enzymes in pigment-formation pathways. Many different geneticists sought to understand the action of genes in terms of their regulation of the rates of chemical reaction, the production of specific chemical products, and the induction of developmental processes. The incredible biochemical complexity of developmental and physiological processes, however, meant that physiological genetics made relatively slow progress.

The gap between genetics and biochemistry was narrowed in 1941, when George Beadle and Edward Tatum began to use the microorganism *Neurospora* to dissect biochemical processes. By growing *Neurospora* on media with different chemical compositions, Beadle and Tatum were able to devise a system for detecting specific changes in the biochemical abilities of their organism. Careful study of biochemical mutants led the two researchers to propose the “one gene–one enzyme” theory, linking genes to specific enzymes and the chemical reactions they catalyzed. Although this association was not new, Beadle and Tatum’s work invigorated the field and encouraged the creation of biochemical genetics as a field of study.

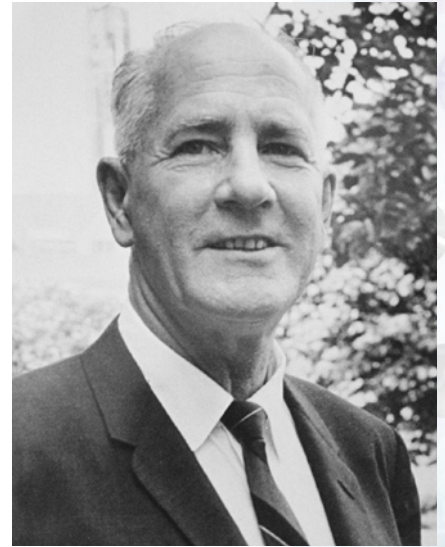
Later it was discovered that many proteins are not enzymes, but instead may be signaling molecules or receptors, or may play structural roles. Thus “one gene–one enzyme” was modified to “one gene–one protein.” It was also discovered that many functional proteins are composed of several distinct amino acid chains (polypeptides) whose corresponding DNA sequences were not necessarily close together or even on the same chromosome, leading to the “one gene–one polypeptide” formulation of the gene definition.

Like the study of gene action, the nature of the gene itself was understood as a biochemical problem early in the twentieth century. Because chromosomes were known to be composed of proteins and nucleic acids, many geneticists proposed specific molecular mechanisms to explain genetic changes or mutations. In 1919 Carl Correns had proposed that the gene was a large molecule with a number of side chains. Mutations were caused by changes in these side chains. Hermann Muller’s pioneering work on the ability of X rays to induce mutations led Nikolay Timofeeff-Ressovsky, Karl Zimmer, and Max Delbrück to investigate the relationship between dose and mutability. The resulting model of mutation, published in 1935, provided a quantum mechanical mechanism for the molecular effects of the ionizing energy of X rays. As the nature and causes of mutation became a more prominent part of genetics, the importance of understanding the molecular basis of the gene also became widely recognized.

## The Molecular Gene

Genetic material was understood in the mid-twentieth century to have two key chemical properties: the ability to **catalyze** reactions to make more genetic material and the ability to catalyze reactions to make a wide array of chemical products found in organisms of all sorts. For most biologists, proteins were the only molecules that seemed to have the ability to play so many specific roles. Researchers naturally focused their attention on the protein component of chromosomes.

The realization that the nucleic acid (DNA) portion of the chromosome was actually the hereditary material was the consequence of two sets of experiments. In 1941 Oswald Avery, Colin MacLeod, and Maclyn McCarty



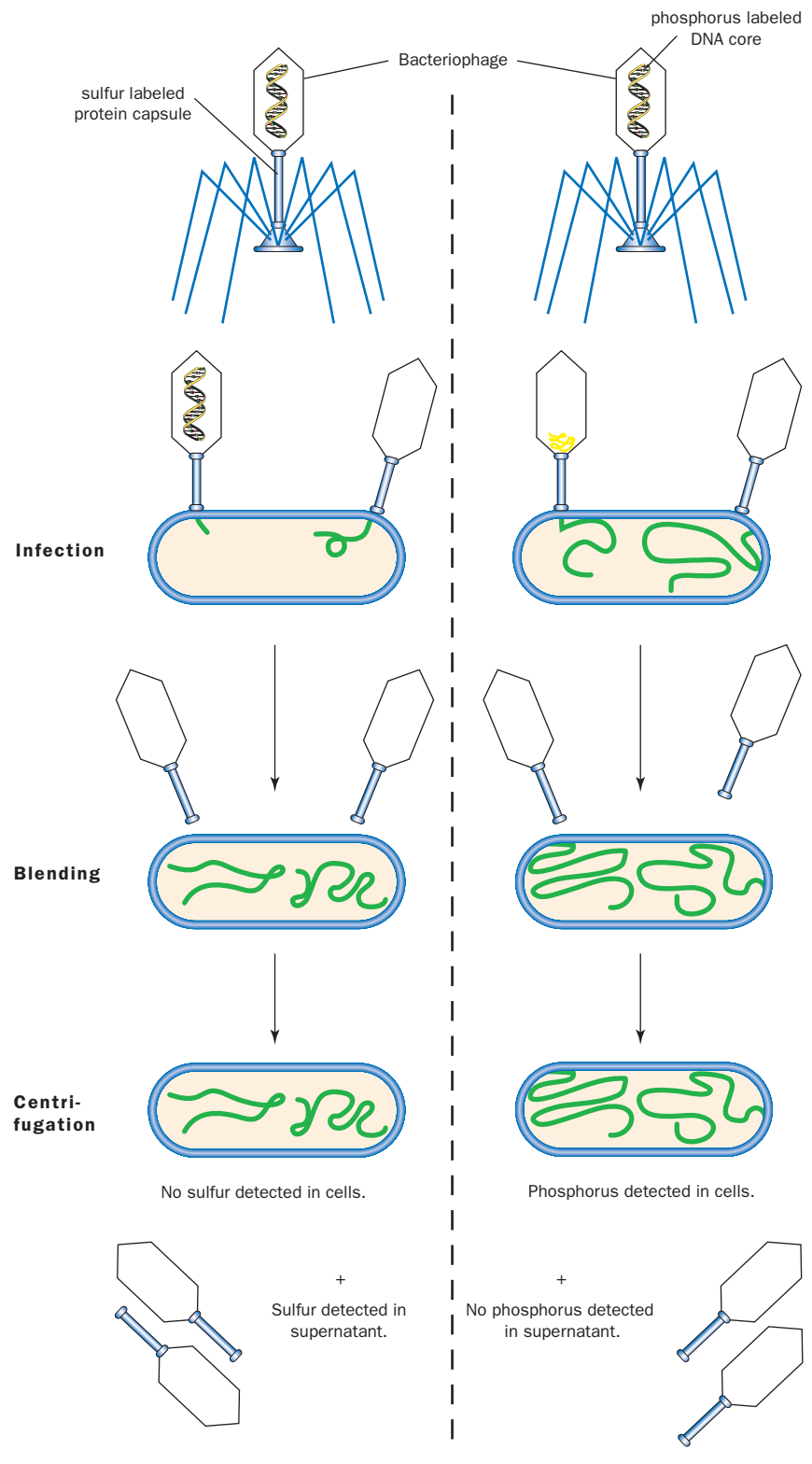
George W. Beadle,  
colleague of Edward  
Tatum.

**enzymes** proteins that  
control a reaction in a  
cell

**catalyze** aid in the  
reaction of

The Hershey-Chase experiment established that DNA, not protein, entered bacteria, and thus was presumably the genetic material. Adapted from <<http://www.accessexcellence.com/AB/GG/HERSHEY.gif>>.

### THE HERSHEY-CHASE EXPERIMENT





extended work by Fred Griffith on the transformation of nonvirulent bacteria into virulent bacteria. Working from the premise that some hereditary chemical component of the virulent bacteria was transforming the nonvirulent bacteria, these researchers isolated DNA and proteins from the virulent bacteria in order to determine which was the “transforming principle.” The surprising result that DNA caused transformation contributed to growing interest in DNA, but DNA was not widely accepted as the genetic material until much later.

In 1952 Alfred Hershey and Martha Chase used radioactive labels to follow DNA and proteins. Hershey and Chase worked with bacteriophages—viruses that infect bacteria. Bacteriophages are composed of proteins and DNA. To determine which was the genetic material, Hershey and Chase created DNA-specific labels and protein-specific labels using radioactivity. They were then able to determine that only DNA was injected into the bacteria to provide the genetic blueprint for the next generation of viruses. This elegant experiment was soon followed by the discovery of the structure of DNA by James Watson and Francis Crick. The double helix structure for DNA immediately suggested a mechanism for its own replication. Thus, by 1953 DNA was identified as having the key catalytic properties required of the genetic material.

At about the same time, the physicist-turned-geneticist Seymour Benzer was using **bacteriophages** to show that genes were not indivisible units; rather, they could break and recombine within their structures. This focused even more attention on the molecular nature of the gene. An understanding of recombination led slowly to a more dynamic view of genes and chromosomes, exemplified by Barbara McClintock’s discovery that some genetic elements are mobile, moving from place to place around the chromosomes. In the early 1960s, Francois Jacob showed that bacterial gene expression is controlled by several noncoding DNA segments. Jacob developed the concept of the operon, a set of coding genes controlled by a common set of regulatory regions.

Identifying DNA as the molecular basis of the genetic material sparked interest in cracking the DNA code and determining how it specifies its products. Reconciling the structure of the DNA sequence with its function became a central preoccupation of molecular genetics.

Ever since Morgan, the gene as a hereditary unit had been a unit of structure and function. Morgan’s particulate gene theory, however, had begun to dissolve in the late 1930s, as it became clear that rearrangements in the chromosome could alter genetic function. Various units of structure and function were suggested (**enzymes**, **polypeptides**, etc.) in the wake of the particulate gene, but the discovery of the genetic code suggested that the molecular gene could be identified as a continuous coding sequence of DNA.

While most coding sequences lead to the formation of a temporary RNA intermediate (messenger RNA) that is then translated into protein, some sequences code for RNA molecules that are not translated but are functional themselves (ribosomal RNA, transfer RNA, and a host of small nuclear RNAs). The discovery of noncoding sequences (introns) within coding regions (exons) further complicated any simple formulation of the structure-function relationship, as did the growing understanding of regulatory

**bacteriophages** viruses that infect bacteria

**enzymes** proteins that control a reaction in a cell

**polypeptides** chains of amino acids



regions, which may or may not be located near the coding regions. Finally, recent discoveries indicate that exons can be joined in different ways in different tissues and that this alternative splicing allows a single set of exons to code for a group of related protein products. By the end of the twentieth century genes could be seen as sequences of DNA (that may be interrupted by noncoding introns) that code for RNA products, many of which are translated into proteins (or a group of related proteins). SEE ALSO ALTERNATIVE SPLICING; CHROMOSOMAL THEORY OF INHERITANCE, HISTORY; DELBRÜCK, MAX; DNA STRUCTURE AND FUNCTION, HISTORY; GENE; INHERITANCE PATTERNS; QUANTITATIVE TRAITS; MEIOSIS; MENDEL, GREGOR; MORGAN, THOMAS HUNT; MULLER, HERMANN.

Michael Dietrich

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### Internet Resource

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## Nomenclature

Like any other field in science, genetics has its own language. However, genetics is also a multidisciplinary field that encompasses expertise, and hence terminology, from diverse areas of science, including molecular biology, statistics, clinical medicine, and, most recently, **bioinformatics**. Despite all of the new and changing language in the field, two of the most frequently used terms in genetics are still “chromosomes” and “genes.”

Humans have twenty-three pairs of chromosomes. One member of each pair is inherited from the person’s mother, and the other from the father. Of the pairs, twenty-two are known as **autosomes**. The remaining pair consists of the sex chromosomes, which determine a person’s gender. Females have two X chromosomes, and males have one X chromosome and one Y chromosome.

Chromosomes are located in the nucleus of a cell. During cell division, which is known as mitosis, the chromosomes’ long strands coil up tightly, to the point where they can be seen as individual units under the microscope. At this stage, each chromosome is composed of two identical strands, called chromatids (each of which further consists of two strands of nucleotides). Chromatids are attached at a constriction point called the centromere.

Chromosomes can be distinguished by their size and by their “banding pattern.” Researchers use a chemical staining process in the laboratory to create the banding pattern, allowing them to see the chromosomes more easily. Each chromosome is divided into two sections, or “arms,” with one arm on each side of the centromere. The short arm is called the p arm, and

**bioinformatics** use of information technology to analyze biological data

**autosomes** chromosomes that are not sex-determining (not X or Y)

the long arm is the q arm. The bands on each arm are numbered. As new and better staining techniques are developed, the numbering system is also refined, so that band 32, for example, would be subdivided into 32.1, then 32.15, and so on. One particular position on the long arm of chromosome 5 would be referred to as 5q32.15.

Almost every human chromosome contains more than a thousand genes. Therefore, even a small extra piece or missing piece of a chromosome results in hundreds of genes being added or deleted from an individual. When researchers study a person's chromosomes, they try to determine if there are any missing or extra chromosomes or chromosome pieces. The addition or deletion of genes sometimes causes a recognizable genetic disorder. Down syndrome, for example, results when there are three copies of chromosome 21 rather than the normal two.

Genes are very small structures that lie on chromosomes. They are the instructions, or blueprints, for producing proteins, which are the building blocks that our bodies use to grow, develop, and function. Humans have an estimated thirty thousand to forty thousand genes.

As happens with other types of scientific discovery, the person who discovers a gene names it, and a scientist can name a gene anything he or she wants. This has led to some confusion, as different naming schemes are used by different groups. To bring order to the situation, several international working groups are trying to standardize the naming of genes. There are separate working groups that focus on naming genes from humans, mice, fruit flies, plants, and other organisms.

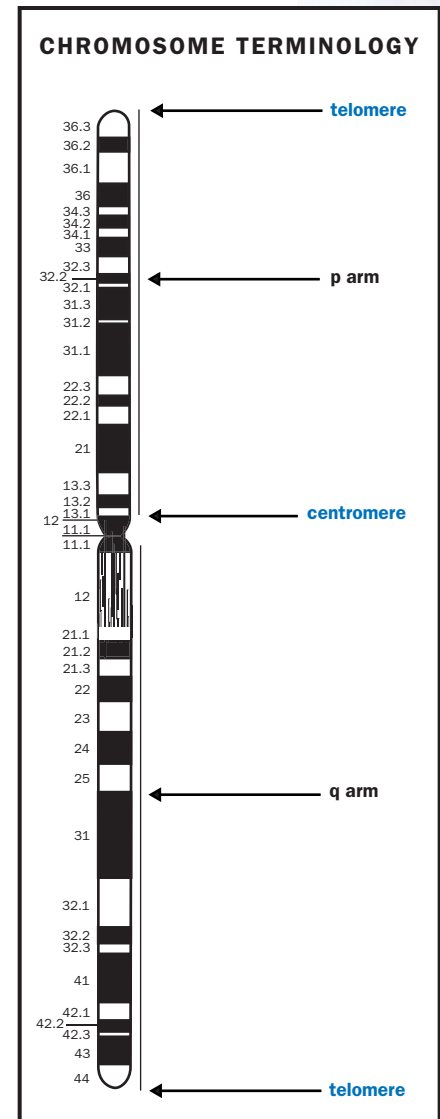
Some scientists choose names based on the clinical disorder that is thought to be associated with changes in the gene. For example, one gene was named *CFTR* because changes in its sequence are associated with the disease cystic fibrosis. Geneticists studying fruit flies traditionally use single-word names, such as *wingless*, *hunchback*, and *sevenless*, that refer to the effect of a mutation in the gene. “Sevenless” refers to the absence of the R7 protein.

For human genes, abbreviations are commonly used. Abbreviated names are especially useful for genes with long names. *WNT2*, for example, stands for “wingless-type MMTV integration site family member 2.” Although the word “wingless” seems unnecessary (humans, of course, don’t have wings!), *WNT2* is named after similar genes in fruit flies. Genes are often named after genes they resemble in other organisms.

Sometimes the gene name is actually a variation of the name of the protein it makes. For example, the *RELN* gene in the human encodes the “reelin” protein. The “reelin” protein was named for the reeling motion exhibited by mice that lack a functional copy of the protein.

Other genes are classified based on what their proteins do. For example, *HOX* genes (short for homeobox) are genes involved in development. Individual *HOX* genes are named with additional letters and numbers, such as *HOXA1* or *HOXD9*. The consistent naming system lets scientists know that any gene with the name *HOX* is likely to play a specific role in development.

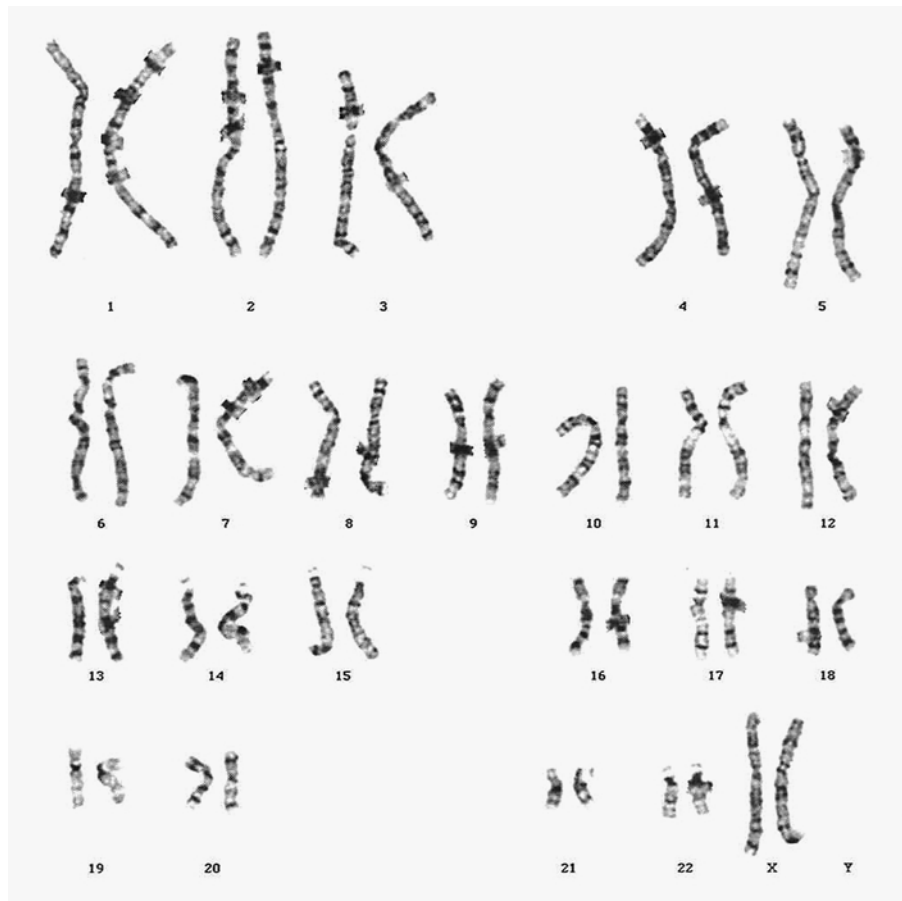
There are even playful gene names that have nothing to do with a disorder or protein. An example is the *SHH* gene, which is involved in the



A chromosome, with banding patterns, band numbers, and major physical landmarks.



This is a photograph of a high-resolution chromosome analysis commonly referred to as a "karyotype." This analysis shows 46, XX which means that this patient has the normal number of chromosomes and is a female since she has two X chromosomes.



development of the brain, spinal cord, and limbs. The *SHH* gene is named after the cartoon character Sonic the Hedgehog! SEE ALSO BIOINFORMATICS; CHROMOSOMAL BANDING; CHROMOSOME, EUKARYOTIC; FRUIT FLY: *DROSOPHILA*.

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### Internet Resource

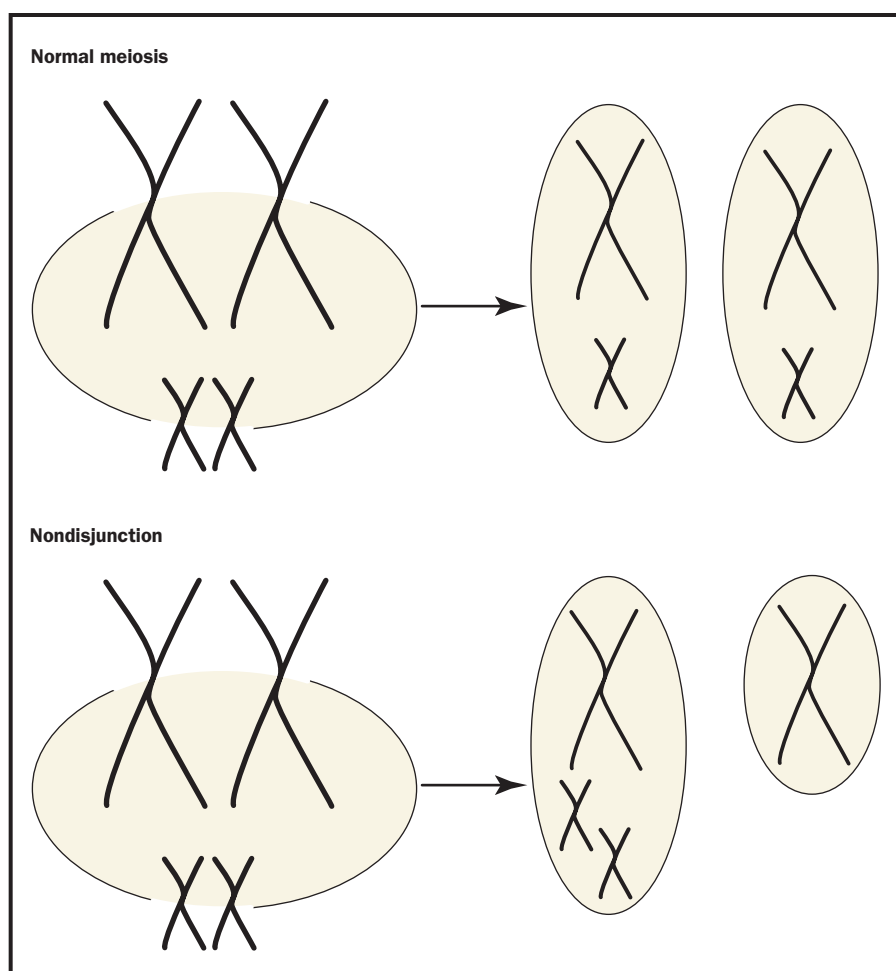
HUGO Gene Nomenclature Committee. <<http://www.gene.ucl.ac.uk/nomenclature>>.

## Nondisjunction

Nondisjunction is the failure of two members of a homologous pair of chromosomes to separate during meiosis. It gives rise to gametes with a chromosomal content that is different from the norm. The consequences of this are usually quite severe, and a number of clinical conditions are the result of this type of chromosome mutation.

### The Mechanism of Nondisjunction

Homologous chromosomes are virtually identical chromosomes that occur in pairs, one member inherited from each parent. Humans have



Nondisjunction occurs when homologous chromosomes fail to separate, creating some cells with too many, and others with too few, chromosomes.

forty-six chromosomes, or twenty-three homologous pairs. In normal meiosis, homologous chromosomes pair up and, by attachment to the spindle fibres, become aligned at the cell equator. Prior to the first meiotic division, the members of each homologous pair migrate to opposite poles of the cell by means of the pulling action of the spindle fibers. This ensures that, upon completion of meiosis, each gamete will contain one copy of every chromosome.

However, the segregation process is not error-free, and every so often it happens that two **homologous** chromosomes fail to separate (disjoin) and thus both migrate to the same pole. This gives rise to two types of gamete. One type possesses two copies of the chromosome, whereas the other type lacks that chromosome altogether. This condition, involving the loss or gain of a single chromosome, is referred to as aneuploidy. Fusion of an aneuploid **gamete** with a normal gamete gives rise to a **zygote** with an odd number of chromosomes.

A zygote which has one less than the normal diploid number of chromosomes ( $2n-1$ ) is said to be monosomic, and such zygotes do not usually develop to term. Zygotes containing an extra chromosome ( $2n+1$ ) are trisomic for the chromosome of interest, and these may develop, though usually with severe abnormalities.

**homologous** carrying similar genes

**gamete** reproductive cell, such as sperm or egg

**zygote** fertilized egg



Various complements of sex chromosomes that yield different phenotypes: two XX, normal female; one X, a Turner syndrome female; one X and one Y, normal male; two X and one Y, a Klinefelter male.



### Non-Fatal Human Aneuploid Conditions

The most common example of non-fatal trisomy in humans is that of Down syndrome, caused by the presence of an extra copy of chromosome 21. Affected individuals suffer from mental retardation, congenital heart disease, and increased susceptibility to infection. Physical characteristics include a short, stocky body, flattened facial features, and almond-shaped eyes.

Down syndrome is an example of an autosomal trisomy as it involves one of the autosomes (“autosome” is the term that designates all the chromosomes other than the X and Y, or sex, chromosomes). There are many human conditions that are caused by nondisjunction of the sex chromosomes, and these usually affect the individual’s secondary sexual characteristics and fertility.

For example, the fusion of an XY sperm with a normal X egg, or the fusion of a Y sperm with an XX egg gives rise to an XXY individual (with normal autosomes). This condition is known as Klinefelter's syndrome. Individuals affected by this disorder usually have below-average intelligence. They are phenotypically male, but present some female secondary sexual characteristics. They may develop breasts, and they have little facial hair, very small testes and are sterile.

Individuals with Turner's syndrome (XO) are females with a single X chromosome. They are sterile and have underdeveloped secondary sexual characteristics, and they are shorter than normal. Females with genetic constitution XXX, on the other hand, have a normal appearance and are fertile, but suffer from a mild mental handicap. Similarly, XYY males have relatively few clinical symptoms and appear phenotypically normal. They are taller than average and may show aggressive behavior and a below-average intelligence. Both XXX and XYY conditions usually pass undiagnosed.

### Fatal versus Nonfatal Conditions

In order to understand why some aneuploid conditions are fatal and others (such as those mentioned above) are not, one must understand the concept of gene dosage and its importance in development. A normal human possesses twenty-two pairs of autosomes and two sex chromosomes (XY in the case of males and XX in the case of females). Such an individual develops normally because there is a situation of genetic balance: Each gene is present in the correct amount (or dose), such that its contribution towards development is appropriate and ideal. However, if a chromosome is either removed from or added to the normal set, a situation of imbalance is immediately established: The contribution (or gene dosage) of each gene contained within that chromosome is altered and as a result development is compromised. While the duplication or silencing of an individual gene is not usually fatal, the wholesale addition or loss of a chromosome, which contains a thousand or more genes, almost always is.

It is obvious from this reasoning that a small change is more likely to be tolerated (albeit at a cost) than a large one. Down syndrome is caused by trisomy of chromosome 21, which is one of the smallest human chromosomes (containing a relatively small number of genes). This provides an explanation as to why this condition is not fatal, while a trisomy involving another, larger autosome would most likely be fatal.

With the sex chromosomes, a lot more flexibility is allowed: Although the X chromosome is very large, only one is used in development (in every female cell one of the two X chromosomes is inactivated at random). The Y chromosome, on the other hand, contains very few genes and is not necessary for normal female development. It is only required for male development. With a knowledge of these facts it is relatively easy to understand why aneuploidies involving the sex chromosomes tend not to be fatal. Note, however, that the YO condition is fatal due to the lack of the essential X chromosome.

### The Causes of Nondisjunction and Its Frequency in Humans

Meiosis is a very tightly regulated process, and a whole series of control mechanisms (constituting a number of "checkpoints") exist to ensure that



everything proceeds in the correct manner. If an error should occur during the process, it is usually corrected. Nondisjunction is the result of a mistake at the level of chromosome segregation, which involves the spindle fibers. In normal meiosis, there is a mechanism that monitors the correct formation of the spindle fibers, the correct attachment of the chromosomes to the spindle fibers, and the correct segregation of chromosomes. Collectively, this is referred to as the spindle checkpoint. Failure of this checkpoint to function correctly results in nondisjunction of chromosomes.

Nondisjunction is known to occur more frequently in the cells of older individuals. This is illustrated by the fact that older women are more likely to give birth to children affected by an aneuploid condition than are younger women. For instance, the risk of a twenty-year-old mother giving birth to a child with Down syndrome is about one in two thousand, compared to an approximate one in thirty risk in the case of a woman of age forty-five. The precise reason for this is not entirely certain, but a simple explanation could be that the older a cell is, the more loosely controlled are the processes occurring within that cell. This would mean that an older cell undergoing meiosis would be more likely than a younger one to ignore the constraints of the spindle checkpoint and hence give rise to aneuploid cells. SEE ALSO CHROMOSOMAL ABERRATIONS; CROSSING OVER; DOWN SYNDROME; MEIOSIS.

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## Nucleases

DNA and RNA are **polymers** made by linking together smaller units called **nucleotides**. Nucleases are enzymes that break the chemical bonds, called **phosphodiester bonds**, that hold the nucleotides of DNA or RNA polymers together. Enzymes that cleave the phosphodiester bonds of DNA are called deoxyribonucleases, and enzymes that cleave the phosphodiester bonds of RNA are called ribonucleases.

Nucleases can be divided into two classes, exonucleases and endonucleases, based on the positions of the cleaved bonds within the DNA or RNA polymers. The exonucleases are involved in trimming the ends of RNA and DNA polymers, cleaving the last phosphodiester bond in a chain. This cleavage results in the removal of a single nucleotide from the polymer. If the enzyme removes nucleotides from the 3' ("three prime") end, it is referred to as a 3' exonuclease. If cleavage is at the 5' end, the enzyme is called a 5' exonuclease. The endonucleases cleave phosphodiester bonds of DNA or RNA at positions other than at the end of the polymer. The cutting reactions of the **endonucleases** produce fragments of DNA or RNA.

Individual nucleases frequently show preferences for various structures of DNA and RNA. Some nucleases prefer single-stranded polymers, while

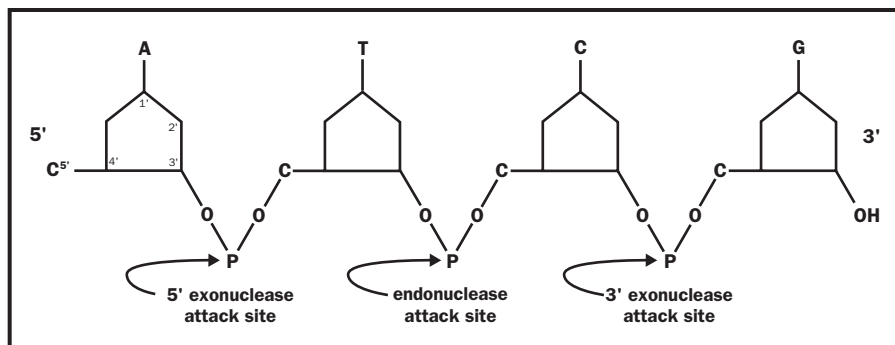
**polymers** molecules composed of many similar parts

**nucleotides** the building blocks of RNA or DNA

**phosphodiester bonds** the links between two nucleotides in DNA or RNA

**endonucleases** enzymes that cut DNA or RNA within the chain





Nucleases are distinguished by their sites of attack. Exonucleases attack the end of a chain, while endonucleases attack intern positions. In reality, the nucleotide polymer could range in length from dozens to thousands of units.

others prefer double-stranded polymers. Some nucleases cleave at specific nucleotide sequences, and others cleave at positions in the polymers independent of nucleotide sequence. Exonucleases can show preference for DNA ends that are correctly base-paired or for ends that are mispaired. Some endonucleases involved in DNA repair recognize damaged nucleotides and **cleave** phosphodiester bonds at these sites. The preferences exhibited by the nucleases reflect the wide biological functions for these **enzymes**.

## The Nuclease Mechanism

There are two different mechanisms used by various nucleases to cleave the chemical bonds of the DNA or RNA polymer. The most common mechanism is one in which a water molecule is used to break the **phosphodiester bond**. This is called a hydrolysis reaction. Under most conditions the P–O (phosphorous and oxygen) bond of the DNA or RNA polymer is very stable, and the H<sub>2</sub>O molecule is not usually very reactive. However, nucleases that use the hydrolysis mechanism make the H<sub>2</sub>O reactive by removing one of the hydrogens to generate a highly reactive OH<sup>−</sup> (hydroxyl). The negatively charged OH<sup>−</sup> can then attack the P–O bond to cleave the polymer. An alternative mechanism used by some DNA repair endonucleases involves the initial cleavage of a C–O (carbon and oxygen) bond and subsequent P–O bond cleavage. This is called a lyase reaction and does not involve water.

## Deoxyribonucleases in DNA Replication and Repair

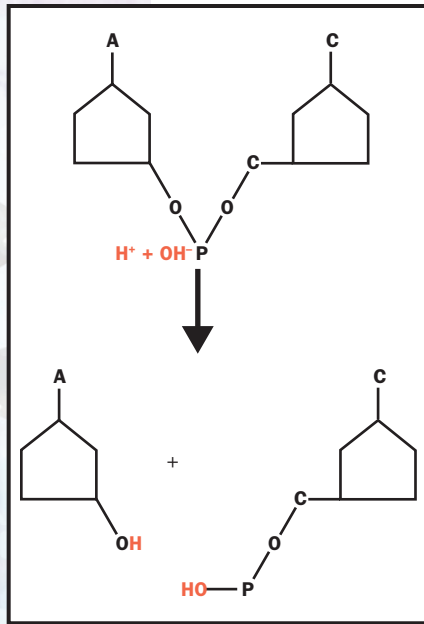
During DNA synthesis the 3′ and 5′ exonucleases function to remove unwanted nucleotides from the DNA. Occasionally, a DNA polymerase will add an incorrect nucleotide to the growing DNA polymer. A 3′ exonuclease removes nucleotides that have been incorrectly polymerized into DNA chains. These exonucleases are referred to as “proofreading” exonucleases. The proofreading exonucleases work in close association with the DNA polymerases to increase the overall accuracy of DNA synthesis.

In many cases the exonuclease activity is contained in the same protein as the DNA polymerase activity. For example, the *Escherichia coli* DNA polymerase I is a single polypeptide with three separate domains, or regions of function. Each of these three domains contains an enzymatic activity. The DNA polymerase activity is in one domain, and the two other domains contain 3′ and 5′ exonuclease activities. The 3′ exonuclease proofreads for the DNA polymerase, and the 5′ exonuclease removes unwanted nucleotides in advance of the DNA polymerase. In contrast, the proofreading exonuclease

**cleave** split

**enzymes** proteins that control a reaction in a cell

**phosphodiester bond** the link between two nucleotides in DNA or RNA



Hydrolysis reaction.

**topological** describes spatial relations, or the study of these relations

Inhalation of recombinant human DNA endonuclease breaks up sticky airway secretions in cystic fibrosis.

of *E. coli* DNA polymerase III is located in a separate protein called the  $\epsilon$  subunit, while the polymerase activity is contained on the  $\alpha$  subunit. These two separate proteins, encoded by different genes, associate and interact in a complex to assure a high level of accuracy during DNA replication.

Multiple DNA repair pathways also use nucleases to restore the correct nucleotide base-pairing if it becomes altered during the life of the cell. Reactive molecules originating from inside the cell during normal metabolism or brought into the cell during exposure to external sources can damage the nucleotides of DNA. A damaged nucleotide opposite a normal nucleotide creates a distortion in the shape of the DNA double helix that is recognized by DNA repair proteins. A DNA helix distortion is also generated when normal but mismatched nucleotides are generated during DNA replication—for example, if a nucleotide is paired with C rather than A. Mismatches occur when DNA polymerases misinsert nucleotides and fail to proofread the misinserted base. These DNA helix distortions are repaired to minimize introduction of mutations into the genome. The steps in these DNA repair pathways include recognition of the distorted DNA, incision of the DNA by endonucleases on the 5' or 3' side of the damage, excision (removal) of nucleotides by exonucleases from the damaged region, and synthesis of a new DNA strand by a DNA polymerase. Some of the genes encoding the repair endonucleases and exonucleases have been identified in *E. coli* and in human cells, and the precise functions of these enzymes in cells are an active area of research.

The topoisomerases are a specialized class of nucleases functioning in cells to alter the **topological** structure of DNA. During replication the DNA becomes twisted, creating a barrier to progression of the DNA replication apparatus. The topoisomerases recognize these twisted regions of DNA and restore them to their untwisted state. This is accomplished by incising the DNA, removing the topological strain by unwinding, then resealing the DNA to regenerate the intact polymer. The Type I topoisomerases function by cutting one of the DNA strands. The Type II topoisomerases cut both DNA strands. The incision of DNA is transient, and both classes of topoisomerases resealed the DNA strands.

The restriction endonucleases are involved in the DNA restriction-modification systems of bacteria, which protect these cells from invading viruses. These enzymes have become powerful tools for DNA manipulation by molecular biologists. They recognize specific sequences in DNA and cut the DNA at these sites. The recognition sequences are usually between four and six nucleotides in length in duplex DNA. Each restriction enzyme has a different recognition sequence, making it possible to cut DNA in a variety of very predictable patterns.

## Ribonucleases in RNA Maturation and Degradation

The expression of genes into protein products requires the generation of a messenger RNA (mRNA) by transcription and the subsequent translation of the mRNA into protein. In bacteria, the mRNA is transcribed, translated, and then degraded by ribonucleases in rapid succession. Thus, the ribonucleases are primarily responsible for mRNA degradation in bacteria. In animal cells, RNA molecules are transcribed as precursors that require processing by ribonucleases to generate functional RNAs. This RNA mat-

uration process requires cleavage by endonucleases and trimming by exonucleases. After the mRNA is translated into protein it is degraded by additional ribonucleases. SEE ALSO CARCINOGENS; DNA POLYMERASES; DNA REPAIR; MUTATION; NUCLEOTIDE; RESTRICTION ENZYMES.

*Fred Perrino*

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## Nucleotide

Nucleotides are the building blocks of deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). Individual nucleotide monomers (single units) are linked together to form polymers, or long chains. DNA chains store genetic information, while RNA chains perform a variety of roles integral to protein synthesis. Individual nucleotides also play important roles in cell metabolism.

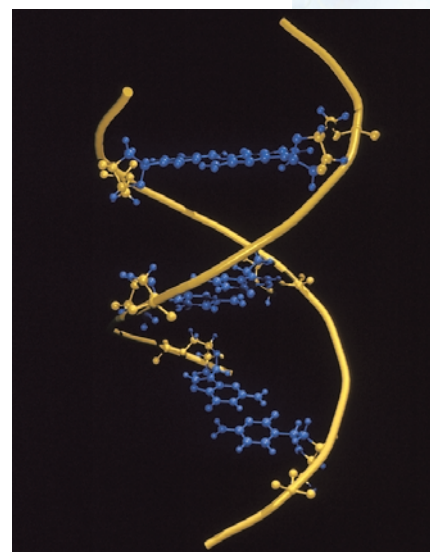
### Structure

The nucleotide molecule contains three functional groups: a base, a sugar, and a phosphate (see diagram). It may seem puzzling that a nucleic acid should contain a base. While the base portion does have weakly basic properties, the nucleotide as a whole acts as an acid, due to the phosphate group.

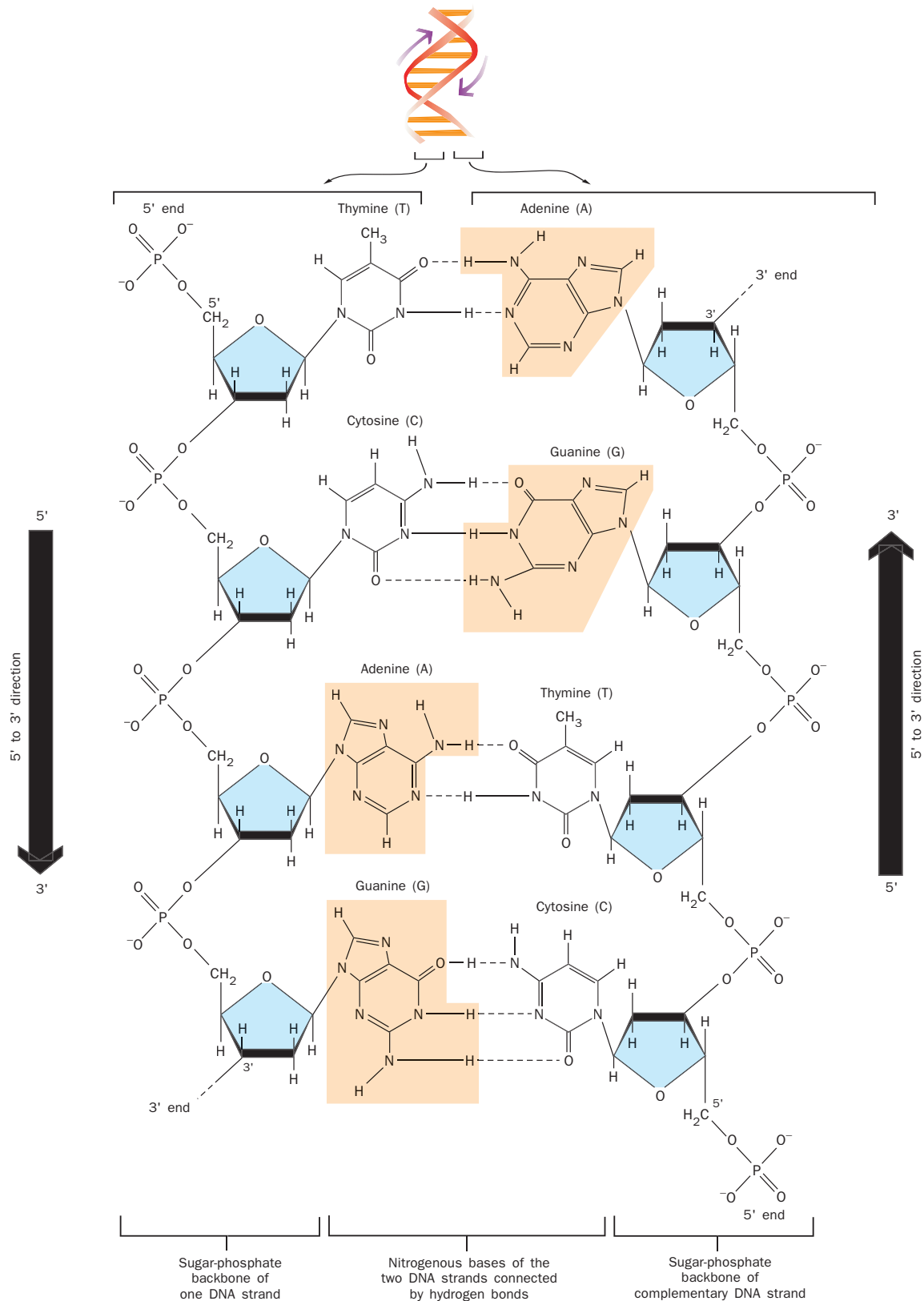
The names DNA and RNA are generated from the deoxyribose and ribose sugars found in these two polymers. Both are five-carbon sugars, whose carbons are numbered around the ring from 1' to 5' ("one prime" to "five prime"). The prime distinguishes the carbons on the sugar from the carbons on the base. The sugar in RNA nucleotides is ribose. The sugar in DNA is 2'-deoxyribose, which lacks an -OH group at the 2' position. This small difference has some important consequences: The extra oxygen in RNA interferes with double helix formation between RNA chains (though it does not completely prevent it), and makes RNA more susceptible than DNA to base-catalyzed cleavage (breakdown into individual monomers).

A base attaches to the sugar at the sugar's 1' position. Because of their nitrogen content, the bases are called nitrogenous bases, and are further classified as either purines or pyrimidines. Purine structures have two rings, while pyrimidines have one. The two purine bases found in both DNA and in RNA are guanine (G) and adenine (A). The two pyrimidine bases found in DNA are cytosine (C) and thymine (T), and the two pyrimidine bases found in RNA are cytosine and uracil (U). The only difference between thymine and uracil is the presence of a methyl group in thymine that is lacking in uracil. A base plus a sugar is called a nucleoside.

The phosphate groups are linked to the sugars at the 5' position. The addition of one to three phosphate groups generates a nucleotide, also known as a nucleoside monophosphate, nucleoside diphosphate, or nucleoside triphosphate. For instance, guanosine triphosphate (GTP) is an RNA



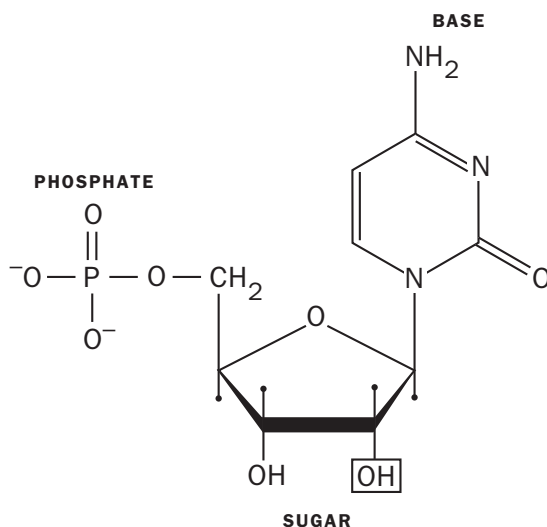
DNA double helix molecular model. The ball-and-stick models linking the backbones represent the nucleotides A, T, C, and G.



DNA nucleotides hydrogen bond across the double helix. A pairs with T, and C with G, due to the specific positioning of the hydrogen bonds (dashes) they form. Each side of the helix has a directionality imposed by the differing characters of the 5' and the 3' ends of the nucleotides. The two sides are oriented in opposite, "anti-parallel" directions.



## NUCLEOTIDES



A nucleotide consists of a nitrogen-containing base, a 5-carbon sugar, and one or more phosphate groups. Carbons occupy four vertices of the pentagon, and are numbered counterclockwise from 1' ("one-prime"), the point of attachment of the base, to 5', the point of attachment of the phosphate. The sugar depicted is ribose; Deoxyribose has an H instead of an OH at the 2' position, indicated by the box.

nucleotide with three phosphates attached. Deoxycytosine monophosphate (dCMP) is a DNA nucleotide with one phosphate attached.

Adenosine triphosphate, ATP, is the universal energy currency of cells. The breakdown of energy-rich nutrients is coupled to ATP synthesis, allowing temporary energy storage and transfer. When the ATP is later broken back down to ADP or AMP (adenosine diphosphate or monophosphate), it provides energy to power cell reactions such as protein synthesis or cell movement.

## Polymer Formation

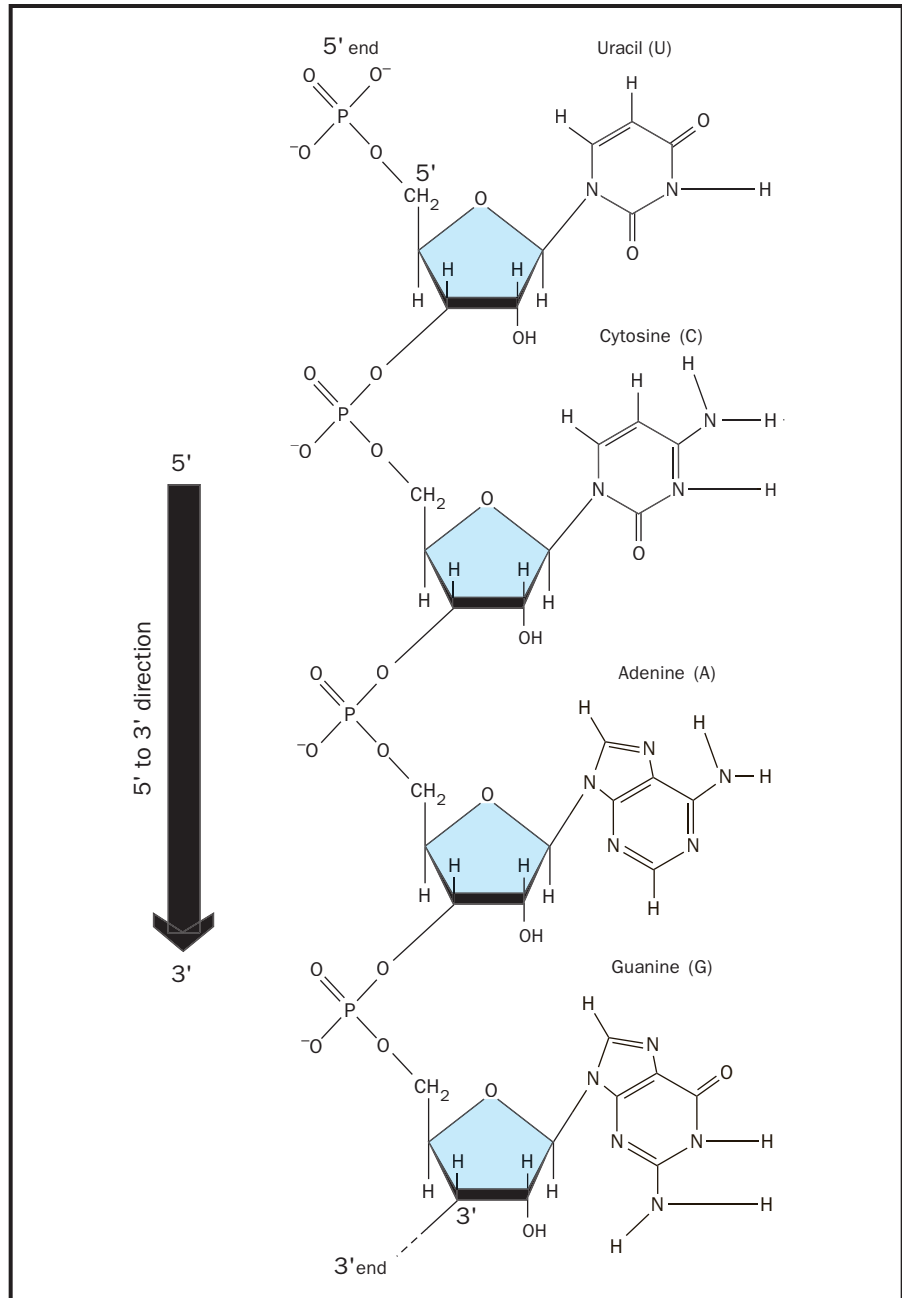
DNA and RNA polymers are constructed by forming **phosphodiester bonds** between nucleotides. In this arrangement, a phosphate group acts as a bridge between the 5' position of one sugar and the 3' position of the next. This arrangement is called the "sugar-phosphate backbone" of DNA or RNA; the bases hang off to the side.

In the cell, DNA or RNA polymers are synthesized using nucleoside triphosphate monomers as precursors. During polymer synthesis, two of the phosphate groups of the incoming nucleoside triphosphate are cleaved off, and this provides the energy needed to power the reaction. The remaining phosphate takes its place in the sugar-phosphate backbone of the growing nucleic acid chain. A pyrophosphate molecule (two linked phosphates) is released.

Just as an arrow has a tip and a tail, DNA or RNA chains have directionality, due to the structure of the sugar. At one end of a chain, a 5' carbon will be left free. This is known as the 5' end of the chain. At the other end, the 3' carbon will be free; this is the 3' end of the chain. Segments of DNA that are not free at their ends can also be discussed in terms of their 5' and 3' ends. This directionality has important consequences. When DNA

**phosphodiester bonds** the links between two nucleotides in DNA or RNA

RNA nucleotides join by linking the 5' phosphate of one to the 3' ribose carbon of the next. This linkage is called a phosphodiester bond.



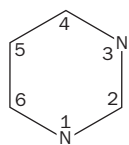
replication occurs, it always moves from the 5' end to the 3' end, and the incoming triphosphate joins the 3' end of the chain. Transcription (RNA synthesis from a DNA gene) also moves in this 5'-to-3' direction. The 5' end is considered the “upstream” end of the gene, and is the end on which the gene promoter (the transcription initiator) is located.

### The Double Helix of DNA

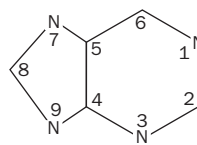
In the double helix of DNA, guanine nucleotides are base-paired opposite cytosine nucleotides. Adenine nucleotides are base-paired opposite thymine nucleotides. This pairing is due to the **complementary** natures of the structures involved. Note that G is a two-ringed purine, while its partner C is a

**complementary** matching opposite, like hand and glove

## BASES

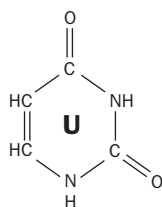


PYRIMIDINE

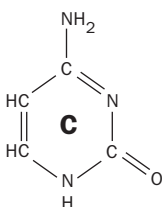


PURINE

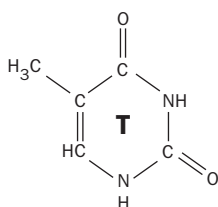
The bases are nitrogen-containing ring compounds, either purines or pyrimidines.



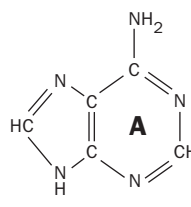
uracil



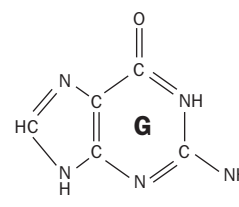
cytosine



thymine



adenine



guanine

one-ringed pyrimidine. Similarly, A is a purine and T is a pyrimidine. These pairings give the interior of the helix a fixed diameter, without bulges or gaps. Just as importantly, the arrangement of atoms in the rings allows the partners to form sets of weak attractions, called **hydrogen bonds**, across the interior of the helix. The hydrogen bonds contribute greatly to the stability of the double helix, and the specificity of the G-C, A-T pairing is the structural basis of faithful replication of DNA. SEE ALSO DNA; DNA POLYMERASE; REPLICATION; RNA; RNA POLYMERASES.

The molecular structures of the five nitrogenous bases.

**hydrogen bonds** weak bonds between the H of one molecule or group and a nitrogen or oxygen of another

*Fred Perrino*

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Stryer, Lubert. *Biochemistry*, 4th ed. New York: W. H. Freeman, 1995.

## Nucleus

The largest of the membrane-bound organelles, the nucleus first was described in 1710 by Antoni van Leeuwenhoek using a simple microscope. In 1831 the Scottish botanist Robert Brown characterized the organelle in detail, calling it the “nucleus,” from the Latin word for “little nut.” The nucleus is the site of gene expression and gene regulation.

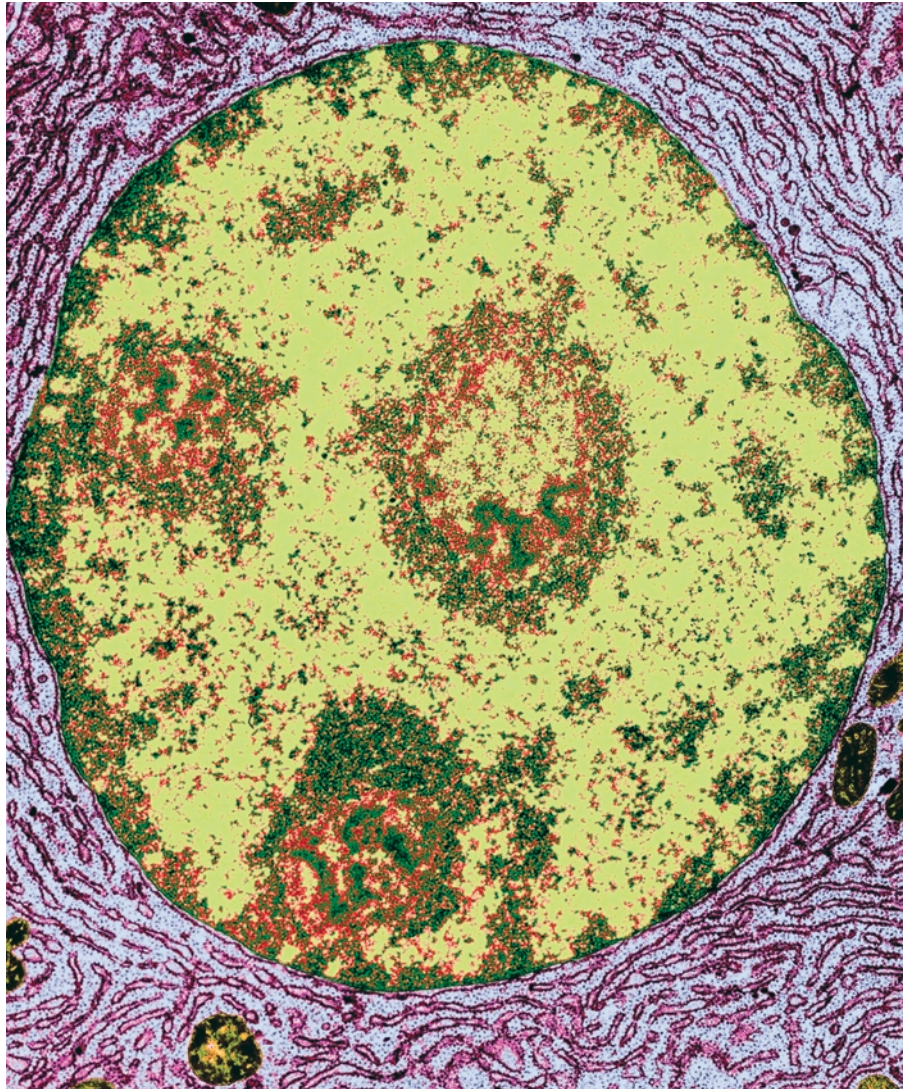
## Distinctive Features

A distinguishing characteristic of eukaryotes, the nucleus contains the genetic information (**genome**) of the cell in the form of its chromosomes. It is within the nucleus that the DNA in the chromosomes is duplicated prior to cell division and where the RNAs are synthesized. Ribosomes are partially assembled around the newly synthesized ribosomal RNAs (rRNA) while still in the nucleus and then transported into the cytoplasm to continue

**genome** the total genetic material in a cell or organism



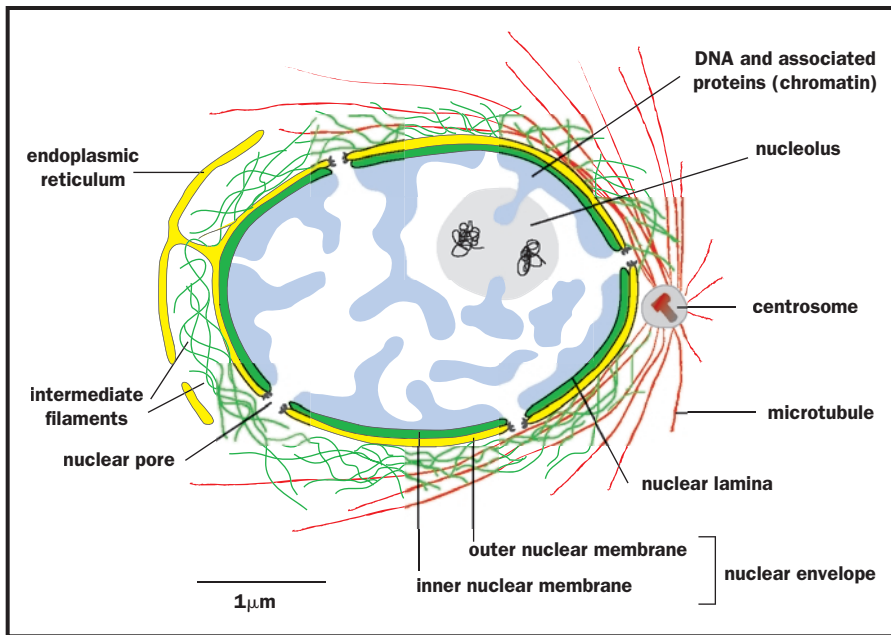
**Figure 1.** Pancreatic cell nucleus magnified nearly 3,000 times by an electron microscope. Stains help to differentiate the various compartments of the nucleus. Note the membrane around the outside, and the extensive endoplasmic reticulum beyond. The dark, stained material within the membrane is chromatin.



their final assembly. Similarly, messenger RNAs (mRNA) are synthesized, packaged, and subsequently transported to the cytoplasmic ribosomes, where they are translated into protein.

Typically spherical in shape and taking up 10 percent of the volume of a cell, the nucleus is bounded by a double membrane called the nuclear envelope (Figures 1 and 2). Most material passes in and out of the nuclear envelope through large openings called the nuclear pores. The outside surface of the envelope is directly connected to the endoplasmic reticulum of the cytoplasm and is surrounded by a network of cytoplasmic intermediate filaments. The inside surface of the nuclear envelope is lined with the nuclear lamina. Internally, the nucleus contains several structures: the chromosomes themselves, which together constitute the chromatin; the interchromatin compartment; the large nucleolus; and a variety of different granules collectively called the subnuclear bodies, which include Cajal (coiled) bodies, gems, PML bodies, and speckles. Every time a cell divides, the nuclear envelope must break down to release the recently duplicated chromosomes. After the chromosomes have segregated to the new daughter cells, the nucleus and its components must be rebuilt.





**Figure 2.** The nucleus holds the cell's DNA. It is surrounded by the nuclear envelope, which is continuous with the endoplasmic reticulum.

If the DNA of each cell were stretched out linearly, it would be over six feet in length. Although the chromosomes of a nucleus appear as a diffuse network in the electron microscope, they are highly compacted into nucleosomal units. Because of nucleosomal folding, the six feet of DNA yields an organelle tightly packed with chromosomal material. Consequently, it was thought that the nucleus in nondividing cells was a fairly static structure, with its various substructures locked into place. Since the 1980s, however, technological advances have permitted investigators to “paint” chromosomes, parts of chromosomes, genes, proteins, RNAs, or subnuclear bodies with genetically defined fluorescent tags. Combined with new techniques that permit these procedures in living cells, and coupled with time-lapse photography and computer simulation, an entirely different image of the cell nucleus is emerging. The nucleus is now understood to be a dynamic organelle composed of a highly ordered architecture that permits a great deal of structural flexibility and movement of molecules and particles between its various subcompartments.

## Chromosomal Territories

Each chromosome is specifically anchored through its **telomeres** to a discrete place on the nuclear envelope by the proteins of the nuclear lamina. Thus each occupies a geographically distinct nuclear space called a chromosomal territory (Figure 2). The homologous chromosomal pairs (matching chromosomes derived from mother and father) do not necessarily lie next to each other.

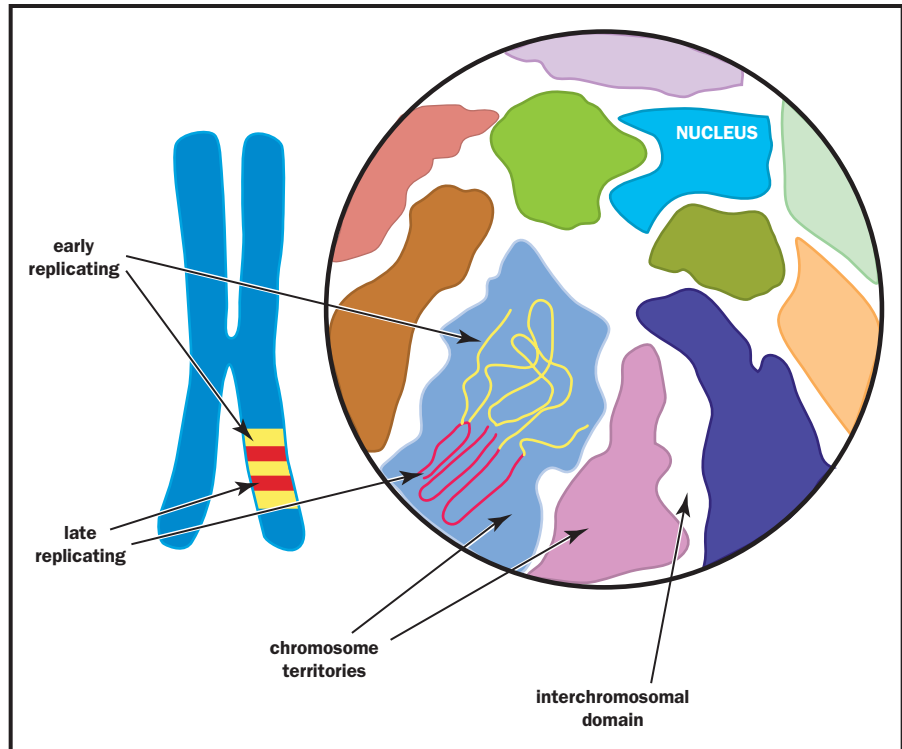
Chromosomal territories are separated by channels of open **nucleoplasm** called the interchromatin compartment. Within each territory, DNA can be highly condensed (heterochromatin) or less condensed (euchromatin). Heterochromatin, defined as DNA that is not currently undergoing active transcription, can contain important chromosomal elements such as **centromeres**. Euchromatin are those chromosomal areas more likely to be active in gene

**telomeres** chromosome tips

**nucleoplasm** material in the nucleus

**centromeres** regions of the chromosome linking chromatids

**Figure 3.** Each chromosome in the nucleus is localized into a territory, separated by more open domains. Early-replicating regions tend to be closer to the center than late-replicating regions.

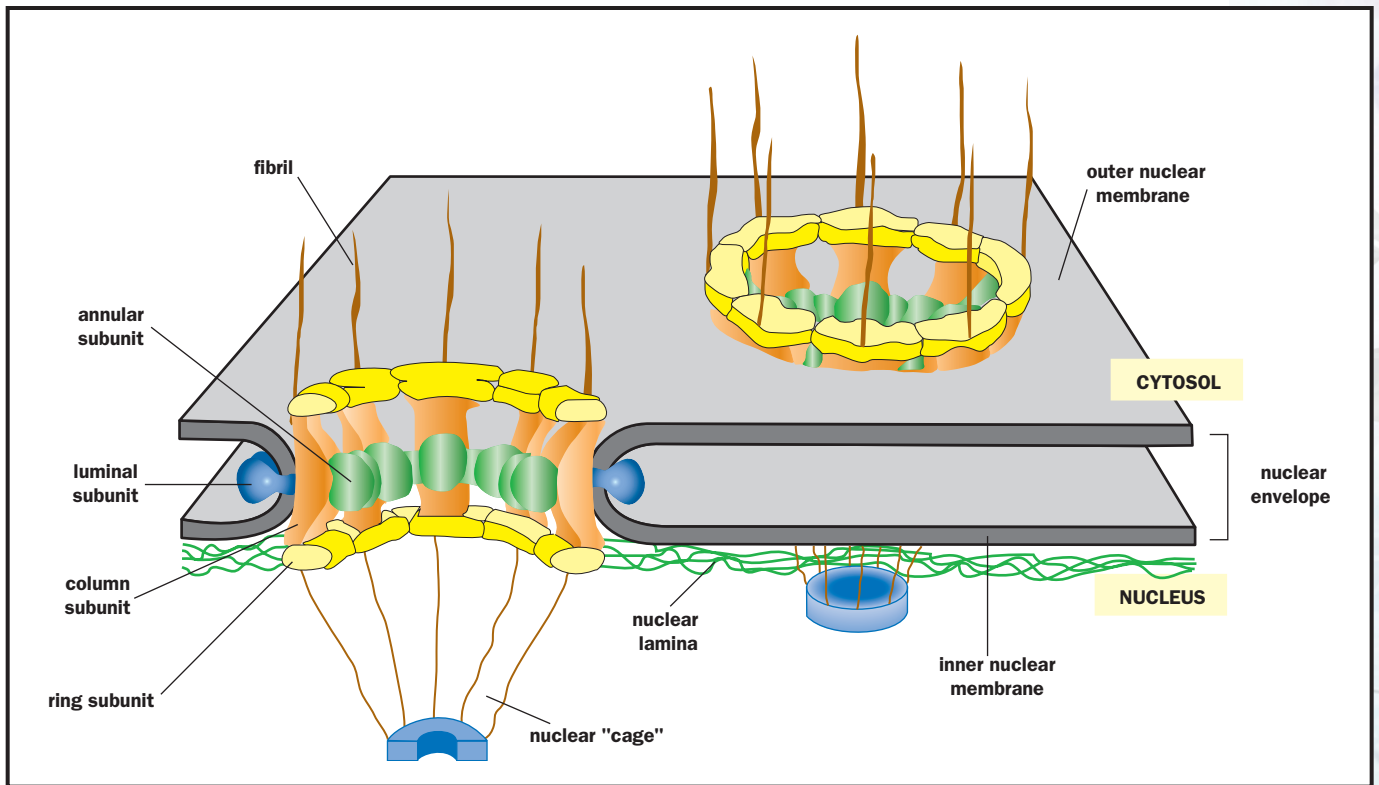


transcription. The heterochromatin of any given chromosome is found within its territory close to the nuclear envelope (Figure 1), but can often project into the interior of the nucleus as patches and/or surround the nucleolus. The euchromatin of each territory extends into the center of the nucleus. In addition, those specific areas of euchromatin undergoing active RNA transcription (gene expression) are typically found on the very periphery of the chromosomal territory, at its juncture with the interchromatin channels.

Chromosomal territories contain at least one other known functional subdomain. Those portions of the DNA that replicate late are found near the nuclear envelope, while earlier-replicating DNA is found in the interior of each territory, projecting into the center of the nucleus. Thus each chromosome not only occupies a discrete place in the nucleus, but each is additionally highly organized into different functional subcompartments. The DNA in each chromosome is highly contorted, looping back and forth within its territory. Chromosomes appear capable of shuffling segments to the correct spot within their territories to carry out gene expression or DNA replication. Indeed, painting of chromosomal segments, including specific genes, with fluorescent tags clearly indicates that chromosomes are constantly shifting around within their territories. Thus the architecture of the chromosomal territories, although highly organized, has a considerable degree of flexibility that is closely tied to both gene expression and DNA replication.

### Interchromatin Compartment

The interchromatin (interchromosomal) compartment is best viewed as a series of channels in and around the individual chromosomal territories that are in direct connection with the nuclear pores of the nuclear envelope. It is filled with nucleoplasm containing subnuclear bodies, nuclear proteins,



**Figure 4.** The double-layered nuclear envelope is studded with nuclear pores, composed of numerous sub-units and related structures.

and RNAs, which move rapidly through its channels. It is thought that as RNA is transcribed from genes along the periphery of the chromosomal territory, it drops into the interchromatin compartment for processing, packaging, and transport out of the nucleus through the nuclear pores.

Hormone receptors, **histones**, and DNA repair enzymes are all known to move actively through these channels, seeking their nuclear targets. Trafficking of molecules is highly efficient; it takes only seconds for a newly synthesized RNA particle to exit through a nuclear pore. Thus the nucleus is a very busy place, with a rapid and continuous exchange of proteins involved in nuclear function and genomic expression occurring both between nuclear compartments and deep within individual compartments, through which access is guaranteed by transportation through the interchromatin compartment.

**histones** proteins around which DNA winds in the chromosome

## Nucleolus

The most prominent nuclear feature, the nucleolus is a ribosomal factory. To make the large number of ribosomes needed, eukaryotic genomes carry multiple rRNA gene copies. The human genome contains 180 rRNA genes located on the tips of five different chromosomes (chromosomes 13, 14, 15, 21, and 22). Anchored by the opposite end to the nuclear envelope, each in their own chromosomal territory, the tip of these five chromosomal pairs (ten chromosomes in a diploid cell) extend into the center of the nucleus and come together, and the rRNA genes align to form what is called the nucleolar organizer. Transcription of the rRNA genes 28S, 18S, and 5S occurs rapidly. The transcripts are immediately processed and sequentially packaged through multiple stages into ribosomal subunits. The processing





**proteomic** derived from the study of the full range of proteins expressed by a living cell

is complicated, requiring many cytoplasmic proteins and enzymes that are transported through the nuclear pores, diffusing through the interchromatin compartment until they reach the nucleolus, where they bind and remain. The nucleolus itself is composed of three subdomains: the nucleolar organizer; rRNA in the process of being transcribed, which is seen as dense fibrils; and granules, which are ribosomes very early in the assembly process.

**Proteomic** analysis indicates that human nucleoli contain at least 271 different proteins of a diverse array of known functions, with 31 percent encoded by unknown genes. This has raised the distinct possibility that the nucleolus performs other functions besides ribosome synthesis. Corroborating data suggest that the nucleolus entraps specific cell-cycle regulatory proteins (such as CDC14), inhibiting their activity until needed. When released from the nucleolus, they regain activity. Nucleoli may also synthesize and/or transport other ribonucleoprotein particles besides the ribosome, and may play a role in the processing and transport of mRNA or tRNA. Because nucleoli are often seen associating with other subnuclear bodies such as Cajal bodies, additional functions are likely.

### Subnuclear Bodies

Although the function of many of the subnuclear bodies remains elusive, they are indeed true nuclear structures. They are seen both by light and electron microscopy and can be studied in living cells through the use of fluorescent tags. They are known to contain complexes of proteins with or without RNA. Like nucleoli, they are not surrounded by a membrane. They often move through interchromatin channels and are thought to represent dynamic complexes that may form and re-form with each other and other nuclear components to process and transport nuclear components.

**Cajal (Coiled) Bodies.** By electron microscopy, Cajal bodies are seen as tangled balls of thread. They number one to ten per nucleus, with more seen in growing cells. They are often found in association with nucleoli or specific chromosomal territories. Although their true function remains unknown, their ability to associate regularly with nucleoli has led to speculation that they are somehow involved in processing either mRNA or rRNA.

**Gems.** More tightly coiled, smaller versions of Cajal bodies, gems are frequently seen interacting with Cajal bodies and are distinct structures. They are known to contain a protein called SMN (which stands for “survival of motor neurons”) that, when mutated, is responsible for a severe inherited form of a human muscular wasting disease called spinal muscular atrophy. Based upon the known function of the normal SMN protein, it is speculated that gems are involved in trafficking mRNA spliceosome subunits through the nucleus and may indirectly help remove mRNA introns.

**PML Bodies.** Nuclei typically have ten to twenty PML bodies (also known as PODs, Kremer bodies, or ND10) that take the shape of dense rings. They contain proteins that, when mutated, have been identified with such disease processes as retinoblastoma and Bloom’s syndrome. Their normal pattern is altered in the nuclei of human acute promyelocytic leukemia. When cells are infected with herpes simplex virus type 1, adenovirus, or human cytomegalovirus, PML bodies are disrupted. Although their function

remains unknown, the fact that they are altered in diseased or malignant cells suggests that they play an important role in the normal cell, including growth control and **apoptosis**.

**Speckles (Interchromatin Granules).** Speckles are clusters of dense structures seen by electron microscopy that, when stained with fluorescent tags specific to small nuclear ribonucleoproteins (snRNP), give rise to a “speckled” nucleus. Small nuclear ribonucleoproteins are RNA-protein complexes that are subunits of the spliceosome involved in mRNA intron removal. The twenty to fifty speckles per nuclei are typically found in the interchromatin compartment, where mRNA undergoes processing prior to transport through the nuclear pore and into the **cytoplasm**.

## Nuclear Envelope

Completely surrounding the nucleus, the nuclear envelope sequesters the genomic information of the cell, probably protecting it from the various enzymes and processes that occur within the cytoplasm. It is composed of two concentric membranes, each of which has a distinct protein composition: the outer membrane, which faces the cytoplasm; and the inner membrane, facing the nuclear interior. The inner and outer membranes are separated by the perinuclear space. Both the outer membrane and the perinuclear space are continuous with the endoplasmic reticulum and studded with ribosomes. Any proteins made on the nuclear outer membrane-bound ribosomes drop into the perinuclear space and are transported through the inner membrane into the nucleus. The major transport pathway in and out of the nucleus, however, is thought to be through nuclear pores.

The inner membrane is coated with a mesh-like network of intermediate filaments called the nuclear lamina. Various nuclear structures, including the chromosomes, attach directly to the lamina, which is essential for maintaining the overall architecture and function of the nucleus. Mutations in the lamina proteins, lamin and emerin, can cause the chromosomes to dissociate from the nuclear envelope and disrupt the organization and properties of the nuclear pores, both of which result in embryonic death. In humans, other lamin mutations cause several rare, inherited diseases, including Emery-Dreifuss muscular dystrophy, an inherited form of muscular dystrophy, or Dunnigan-type lipodystrophy, a disease that results in loss of adipose tissue and late-onset, insulin-resistant diabetes beginning at puberty. How lamina protein mutations cause these two diseases is unknown.

## Nuclear Pores

Perhaps the most startling feature of the nuclear envelope are the very large, basket-like transport structures called the nuclear pores (figure 4). These structures have a molecular weight of 125 million daltons, making them thirty times larger than a ribosome. Composed of 100 to 200 different proteins collectively called nucleoporins, each nuclear pore pierces through both membranes of the nuclear envelope and probably opens into the interchromatin space of the nucleus. Some nucleoporins are structural components of the nuclear pore; others facilitate transport. Each mammalian cell nucleus contains 3,000 to 5,000 of these pores. The large number is needed to transport the tremendous quantity of proteins, enzymes, RNAs,

**apoptosis** programmed cell death

**cytoplasm** the material in a cell, excluding the nucleus

